

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0976

Stuttgart Ricuskey 100 kV
May 31, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0976. The Interconnection Customer proposes a 201.1 MW natural gas combined cycle facility interconnecting at the Stuttgart Ricuskey 100 kV substation in the Illinois area. The request was received on April 05, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 399 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,194,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TROY ENERGY LLC - DIXON LANDING 230 kV line	TROY ENERGY LLC SUBSTATION - RICHLAND 500 kV	199.0	558	8.6
E METHUEN 500 kV/230 kV Transformer Bank	TREGO - BEAVER 500 kV	122.6	1,605	16.0
MADELINE - BARAGA VLG 230 kV	GRAY'S FLAT SUBSTATION - EAST MEHUEN SUBSTATION 230 kV	116.3	514	6.5
ADIRONDACK - JAMES DE YOUNG 230 kV	ST. LAWRENCE SUBSTATION - JAMES DE YOUNG 230 kV	115.7	748	7.4
CORRIDOR - LARSON ST 230 kV	TREGO - BEAVER SUBSTATION 500 kV	114.4	640	9.9
ADIRONDACK SUBSTATION 500-230 kV Auto	E METHUEN 500 kV/230 kV Transformer Bank	113.3	1,026	5.1
WALDO 500/230 kV TX	TREGO - BEAVER 500 kV	111.0	894	9.8
CARLEY RD - JAMES DE YOUNG 230 kV	E METHUEN 500-230 kV Autotransformer	110.2	647	7.1
TREGO 230 kV/1 kV Transformer Bank	TREGO - TROY ENERGY LLC 500 kV Line	109.9	644	12.9
JOSO - ONANCOCK DIESEL GENERATION FACILITY 230 kV Line	BEAVER - EUFAULA SUBSTATION 500 kV	108.9	1,738	14.3
ADAMS - PICKFORD 230 kV	TREGO - BEAVER 500 kV	108.7	1,089	8.1
ADAMS - ONANCOCK DIESEL GENERATION FACILITY 230 kV	TREGO - BEAVER 500 kV	108.4	2,118	15.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PICKFORD - HERALD-TWIN CITIES 230 kV	TREGO - BEAVER SUBSTATION 500 kV	106.7	1,423	9.7
MASON ST - NEW DEPARTURE 230 kV	TREGO - BEAVER 500 kV Line	105.7	1,109	7.8
TREGO - HIGHGATE 230 kV	TREGO - TROY ENERGY LLC 500 kV	105.6	511	10.9
HEBO - WINTHROP 230 kV Line	BARRETT - HEBO 230 kV	105.2	629	5.5
BURRIS PIT - HEBO 230 kV	TREGO - BEAVER 500 kV	101.4	1,021	7.9
JAMES DE YOUNG - BURRIS PIT 230 kV	BEAVER - TREGO SUBSTATION 500 kV	101.1	1,046	7.9
MADLIE - E METHUEN 230 kV	GRAY'S FLAT - EAST METHUEN 230 kV	100.8	1,074	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DIXON LANDING - TROY ENERGY LLC 230 kV L	Rebuild line with higher-capacity conductor	\$23,706,000	2% (\$404,000)
E METHUEN 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$27,788,000	2% (\$467,000)
MADLINE - BARAGA VLG 230 kV line	Rebuild line with higher-capacity conductor	\$10,784,000	2% (\$181,000)
ADIRONDACK - JAMES DE YOUNG 230 kV	Rebuild line with higher-capacity conductor	\$53,370,000	2% (\$896,000)
LARSON STREET - CORRIDOR 230 kV	Rebuild line with higher-capacity conductor	\$18,521,000	2% (\$313,000)
ADIRONDACK SUBSTATION 500-230 kV Auto	Replace transformer with higher-rated unit	\$18,489,000	2% (\$310,000)
WALDO 500/230kV TX	Replace transformer with higher-rated unit	\$31,085,000	2% (\$524,000)
CARLEY RD - JAMES DE YOUNG 230 kV	Rebuild line with higher-capacity conductor	\$31,111,000	2% (\$522,000)
TREGO SUBSTATION 230/1 kV TX	Replace transformer with higher-rated unit	\$8,992,000	2% (\$153,000)
JOSO - ONANCOCK DIESEL GENERATION FACILITY 230 kV	Reconductor line	\$5,442,000	2% (\$92,000)
PICKFORD - ADAMS 230 kV	Reconductor line	\$8,860,000	2% (\$149,000)
ADAMS - ONANCOCK DIESEL GENERATION FACILITY 230 kV line	Reconductor line	\$21,767,000	2% (\$366,000)
PICKFORD - HERALD-TWIN CITIES SUBSTATION 230 kV	Reconductor line	\$24,783,000	2% (\$417,000)
MASON ST - NEW DEPARTURE 230 kV	Reconductor line	\$12,779,000	2% (\$215,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HIGHGATE - TREGO 230 kV	Reconductor line	\$9,867,000	2% (\$169,000)
HEBO - WINTHROP 230 kV Line	Reconductor line	\$4,926,000	2% (\$83,000)
BURRIS PIT - HEBO 230 kV	Reconductor line	\$6,197,000	2% (\$104,000)
BURRIS PIT - JAMES DE YOUNG 230 kV L	Reconductor line	\$36,425,000	2% (\$611,000)
E METHUEN - MADELINE 230 kV	Reconductor line	\$13,004,000	2% (\$218,000)
	Total allocated to project		\$6,194,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.